

The Effects of Phytosterols Present in Natural Food Matrices on Cholesterol Metabolism and LDL-Cholesterol: A Controlled Feeding Trial

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Background/Objectives Extrinsic phytosterols supplemented to the diet reduce intestinal cholesterol absorption and plasma LDL-cholesterol. However, little is known about their effects on cholesterol metabolism when given in native, unpurified form and in amounts achievable in the diet. The objective of this investigation was to test the hypothesis that intrinsic phytosterols present in unmodified foods alter whole-body cholesterol metabolism. Subjects/Methods Twenty out of 24 subjects completed a randomized, crossover feeding trial where all meals were provided by a metabolic kitchen. Each subject consumed two diets for 4 weeks each. The diets differed in phytosterol content (phytosterol-poor diet, 126 mg phytosterols/2000 kcal; phytosterol-abundant diet, 449 mg/2000 kcal) but were otherwise matched for nutrient content. Cholesterol absorption and excretion were determined by gas chromatograph/mass spectrometry after oral administration of stable isotopic tracers. Results The phytosterol-abundant diet resulted in lower cholesterol absorption [54.2 ± 2.2 % (95% confidence interval, 50.5%, 57.9%) vs. 73.2 ± 1.3 % (69.5%, 76.9%), $P < 0.0001$] and 79% higher fecal cholesterol excretion [1322 ± 112 (1083.2, 1483.3) vs. 739 ± 97 mg/day (530.1, 930.2), $P < 0.0001$] relative to the phytosterol-poor diet. Plasma lathosterol/cholesterol ratio rose 82% [from 0.71 ± 0.11 (0.41, 0.96) to 1.29 ± 0.14 μ g/mg (0.98, 1.53), ($P < 0.0001$)]. LDL-cholesterol was similar between diets. Conclusions Intrinsic phytosterols at levels present in a healthy diet are biologically active and have large effects on whole body cholesterol metabolism not reflected in circulating LDL. More work is needed to assess the effects of phytosterol-mediated fecal cholesterol excretion on coronary heart disease risk in humans.